ALOK RAM TRIPATHI

FULL-STACK SOFTWARE ENGINEER · AWS CERTIFIED DEVELOPER (ASSOCIATE)

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PROFESSIONAL SUMMARY

Full-stack software engineer and AWS Certified Developer (Associate) with over a year of professional experience architecting and shipping production web applications used by 5,500+ people. Takes full ownership of features end to end: SEO-optimized React and Next.js frontends, REST APIs engineered in Python and Node.js, and PostgreSQL data models for real business workflows, deployed as containerized services on AWS with Docker and CI/CD.

PROFESSIONAL EXPERIENCE

Software Engineer · MandS Consulting May 2025 – Present

- Drive end-to-end ownership of features for two client products, Braided (1,500+ users) and Hey Coach (4,000+ users, growing monthly), from design review to production release.
- Built a production RAG pipeline on Braided using OpenAI embeddings and a pgvector store, with semantic chunking and similarity search to ground LLM responses in product data, cut hallucinations, and stream cited answers to users.
- Engineer SEO-optimized, server-rendered interfaces in Next.js and React, strengthening search crawlability and cutting page-load times on high-traffic product pages.
- Architected a shared library of reusable, responsive Tailwind CSS components that accelerates feature delivery and enforces one design system across both products.
- Architected and optimized PostgreSQL schemas, indexes, and queries powering complex business workflows, driving faster query responses and stronger reliability as usage scales.
- Engineer backend services and REST APIs in Python that power data exchange across the frontend, backend, and database layers for 5,500+ combined users.
- Delivered a cross-platform mobile app in Flutter with a responsive UI and full REST API integration.
- Overhaul legacy modules to eliminate technical debt and accelerate page loads, partnering with product and design to scope and ship requirements on time.

MERN Stack Developer Intern · Vector India Pvt. Ltd. Jun 2024 – Mar 2025

- Engineered full-stack applications with MongoDB, Express.js, React.js, and Node.js, implementing JWT authentication, CRUD REST APIs, and client-side routing.
- Shipped applications to Vercel and Render through Git and GitHub workflows, contributing to peer code reviews and clean-code standards.

TECHNICAL SKILLS

LANGUAGES	JavaScript, TypeScript, Python, SQL, HTML5, CSS3
FRONTEND	React.js, Next.js, Tailwind CSS, Bootstrap, Flutter, Responsive and SEO-Optimized UI
BACKEND	Node.js, Express.js, REST APIs, JWT Authentication
AI / GENAI	RAG Pipelines, Vector Databases (pgvector), Embeddings, Semantic Search, LLM APIs, OpenAI, LLM Integration
DATABASES	PostgreSQL, MySQL, MongoDB
CLOUD, DEVOPS	AWS (ECS, ECR, EC2, VPC, Route 53, ALB, IAM, CodePipeline, CodeBuild), Docker, CI/CD
TOOLS	Git, GitHub, Bitbucket, Jira, Postman, Agile/Scrum, Code Review

CERTIFICATIONS

- AWS Certified Developer – Associate · Amazon Web Services
- Frontend Developer (React) · HackerRank, Mar 2025 · ID F7B45AE4CB72
- MERN Stack Full-Stack Web Development Training · Vector India Pvt. Ltd.
- Web Design Internship, Certificate of Completion · Universal IT Solution
- AI/ML for Geodata Analysis, Participation Certificate · ISRO

EDUCATION

B.Tech, Information Technology · Mahakal Institute of Technology Aug 2019 – May 2023

CGPA 7.50 / 10 Ujjain, Madhya Pradesh